DSLL-Net: A Dual-Stage Leaf–Lesion Segmentation Framework for Automated Plant Disease Severity Quantification

Changqing Yan[a], Shuyang Li[a], Zhe Liu[a], Ling Yin [b], Shumei Wei [b], Xin Lin[a], Xin Li[a], Jindong Liu[a], Zhenpeng Huang[a], Danni Zheng[a], Zeyun Liang[a], Qi Tian[c], Linjia Yao[c], Bin Chen [d], Chenliang Wang[e], Junjie Qu[b]*, Gang Zhao [d]*

[a]College of Intelligent Equipment, Shandong University of Science and Technology, Taian 271019, China

[b]Guangxi Crop Genetic Improvement and Biotechnology Key Lab, Guangxi Academy of Agricultural Sciences, Nanning 530007, China

[c]College of Natural Resources and Environment, Northwest A & F University, Yangling 712100, China

[d]State Key Laboratory of Soil and Water Conservation and Desertification Control, College of Soil and Water Conservation Science and Engineering, Northwest A&F University, Yangling 712100, China

[e]School of Data Science and Artificial Intelligence, Wenzhou University of Technology, Wenzhou 325000, China

*Corresponding author: Gang Zhao, E-mail address: gang.zhao@nwafu.edu.cn; Junjie Qu, Email address: qujunjie@gxaas.net

Abstract

Quantitative disease severity assessment from field images requires accurate estimation of both leaf and lesion areas, yet complex backgrounds, illumination variation, occlusion, and small lesion patterns remain challenging. This study proposes DSLL-Net, a lightweight dual-stage leaf–lesion segmentation framework for grapevine downy mildew severity quantification. DSLL-Net first segments leaves using DeepLabV3+ with a SimAM-enhanced MobileNetV3 backbone, and then delineates lesions with an improved U-Net integrating a SimAM-EfficientNet-B0 encoder and a RepConv decoder. This task decomposition reduces feature interference between leaf boundaries and disease symptoms while maintaining computational efficiency. Experiments on field datasets show that DSLL-Net achieves a favorable accuracy–efficiency trade-off, with a lesion IoU of 74.3% using 12.56 M parameters. Severity estimates derived from segmented leaf and lesion areas strongly agree with reference values ($R^2 = 0.976$). The framework was further integrated into an Android application, demonstrating its potential for field-oriented disease monitoring and decision support.

Introduction

Crop diseases threaten agricultural production and require accurate severity assessment for disease management, fungicide evaluation, yield loss estimation, and resistance screening [1-5]. Conventional visual assessment is subjective and difficult to reproduce [6-8]. Although image-level classification methods can assign disease samples to predefined severity grades [9], they provide coarse estimates and depend heavily on manually defined categories.

Pixel-wise semantic segmentation enables more quantitative severity estimation by calculating lesion coverage on leaf regions [10]. Recent segmentation methods have improved disease detection through lightweight architectures and attention mechanisms [11, 14-17]. However, many approaches segment leaves and lesions in a single network. Because severity is calculated as the ratio of lesion area to leaf area, errors in either component directly affect the final estimate. In complex field images, single-stage segmentation can suffer from feature interference between large leaf boundaries and small, irregular lesions, leading to unstable severity quantification.

Dual-stage frameworks reduce this interference by separating leaf extraction from lesion delineation. Existing methods such as DUNet, LD-DeepLabv3+, StripeRustNet, and CL-VW-DeepLabV3+ have demonstrated the benefit of task decomposition for disease severity estimation. However, field deployment still requires models that balance segmentation accuracy, compactness, inference efficiency, and usability on resource-constrained devices.

In this study, we propose DSLL-Net, a lightweight dual-stage framework for quantitative grapevine downy mildew severity assessment from field images. The main contributions are: (1) a task-decoupled leaf–lesion segmentation pipeline that reduces interference between leaf boundaries and small lesions; (2) a compact leaf segmentation model based on SimAM-MobileNetV3 and DeepLabV3+ and an efficient lesion segmentation model based on SimAM-EfficientNet-B0 and RepConv U-Net; and (3) a severity estimation and mobile deployment workflow validated under field conditions.

Materials and Methods

Overview of the workflow

Fig. 1 summarizes the proposed workflow. Field images were annotated to construct task-specific datasets for leaf segmentation, lesion segmentation, and single-stage comparison. DSLL-Net then performs severity estimation in two stages: SimV3-DL3+ first extracts leaf regions from complex backgrounds, and SimEffB0-RepUNet subsequently segments lesions within the leaf region. Disease severity is calculated as the ratio of lesion area to total leaf area and is integrated into a mobile application workflow.

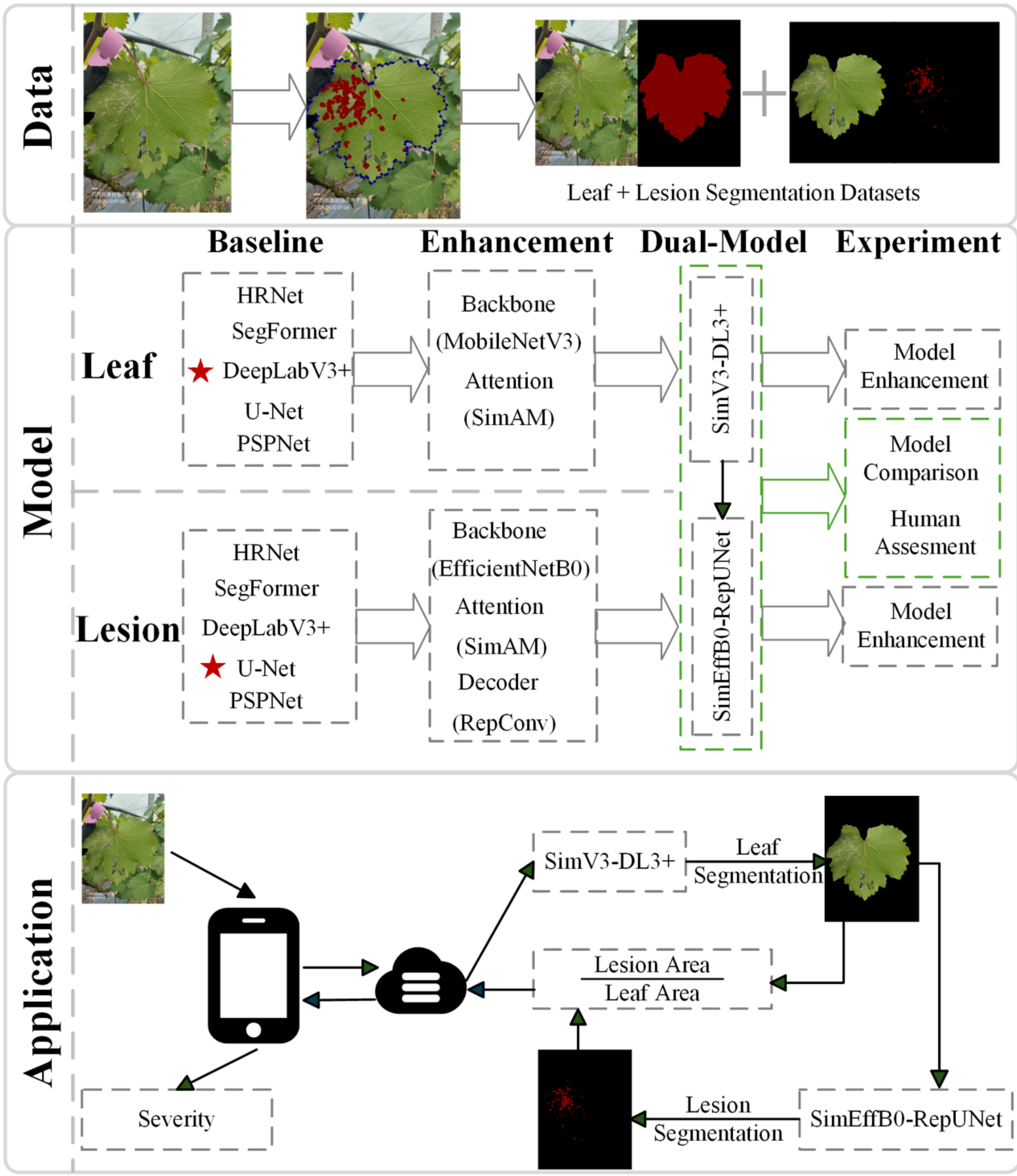

Fig. 1. Overall workflow of data preparation, dual-stage model development, and mobile application deployment.

Data Preparation

A total of 696 field images were collected between April and August 2024 at Mingyang and Du'an in Guangxi, China. Images were captured using a Huawei P70 and an iPhone 15 at a distance of 15–20 cm, with each image focused on a single leaf. Images with clear leaf structures, appropriate viewing angles, and no severe occlusion were retained.

Leaf and lesion regions were manually annotated using LabelMe (v3.16.7), generating binary masks for leaf/background and lesion/non-lesion classes. These annotations were used to construct three datasets for leaf segmentation, lesion segmentation, and single-stage lesion segmentation, as summarized in Table 1.

Table. 1 An overview of the constructed datasets and their corresponding modeling tasks.

Dataset	Input to model	Label(s)	Co
Leaf segmentation	Original full RGB field image	Leaf mask (binary)	Ma
Lesion segmentation	Cropped leaf ROI	Lesion mask (binary, cropped consistently)	Cro
Single-stage segmentation	Original full RGB field image	Lesion mask (binary)	Ori

All datasets used the same train–test split: 567 images for training and 129 for testing. To match deployment conditions, the second-stage lesion model used leaf regions predicted by the first-stage model rather than ground-truth leaf masks.

Dual-Stage Segmentation Model

Leaf Segmentation Model SimV3-DL3+

The first stage aims to segment leaf regions accurately while keeping the model compact for mobile deployment. Five representative segmentation models (DeepLabV3+, U-Net, PSPNet, SegFormer, and HRNet) were evaluated, and DeepLabV3+ was selected as the baseline because it provided the best accuracy with moderate complexity. To improve efficiency, MobileNetV3 was used as the backbone. Its original SE module was replaced with SimAM, which enhances spatial responses without adding learnable parameters. The resulting SimV3-DL3+ model is shown in Fig. 2.

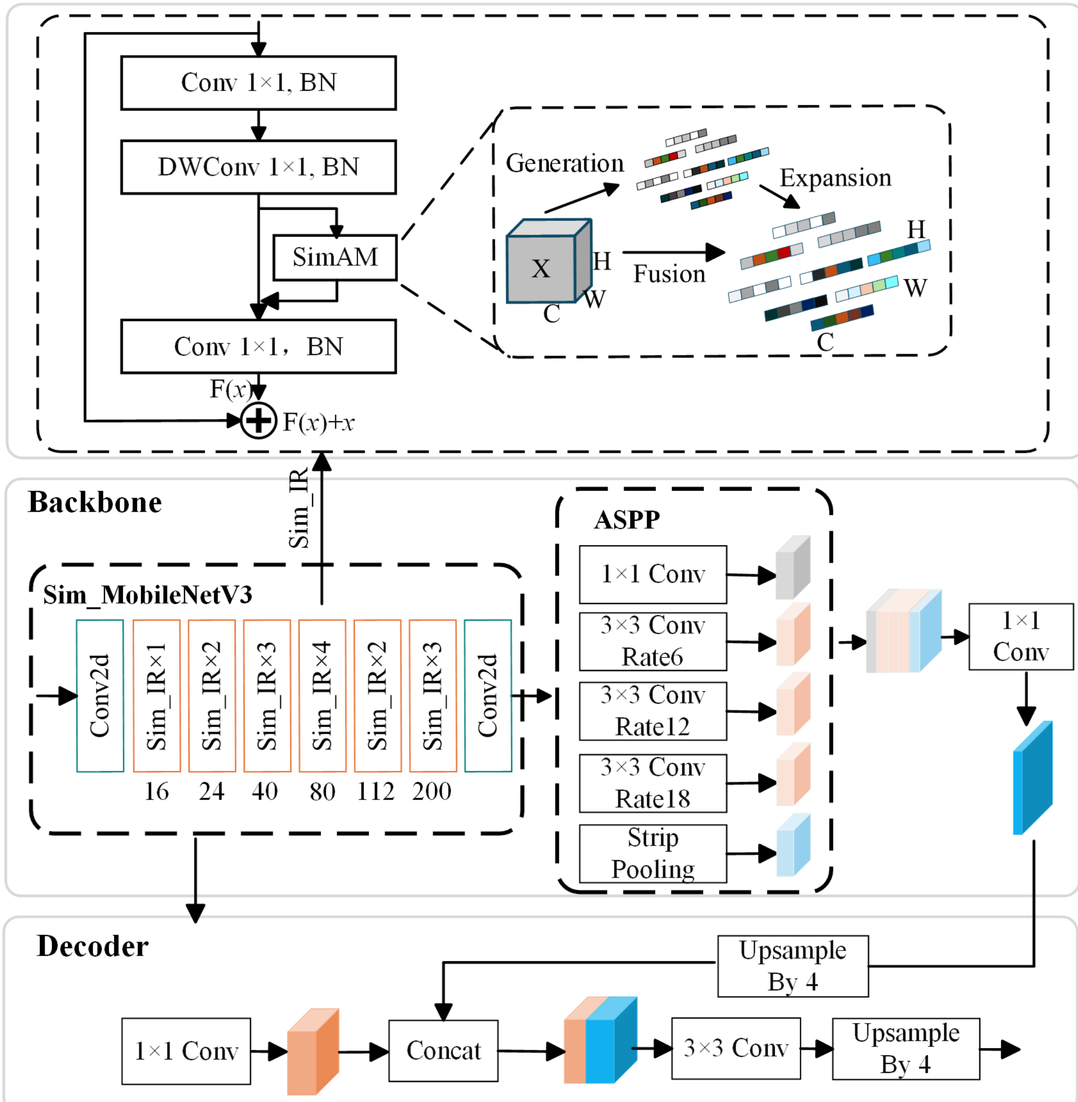

Fig. 2. Architecture of the SimV3-DL3+ model. Sim_IR denotes the InvertedResidual block enhanced with SimAM attention.

Lesion Segmentation Model (SimEffB0-RepUNet)

The second stage focuses on small and irregular lesion regions, where boundary precision is more difficult than leaf extraction. U-Net was selected as the baseline after comparison with DeepLabV3+, PSPNet, SegFormer, and HRNet. To reduce the parameter cost of the conventional VGG16 encoder, EfficientNet-B0 was adopted as the encoder. Its SE module was replaced with SimAM to improve spatial lesion representation without increasing model complexity. RepConv blocks were further introduced in the decoder to strengthen feature fusion and boundary reconstruction during training while preserving efficient inference after re-parameterization. The final model, SimEffB0-RepUNet, is shown in Fig. 3.

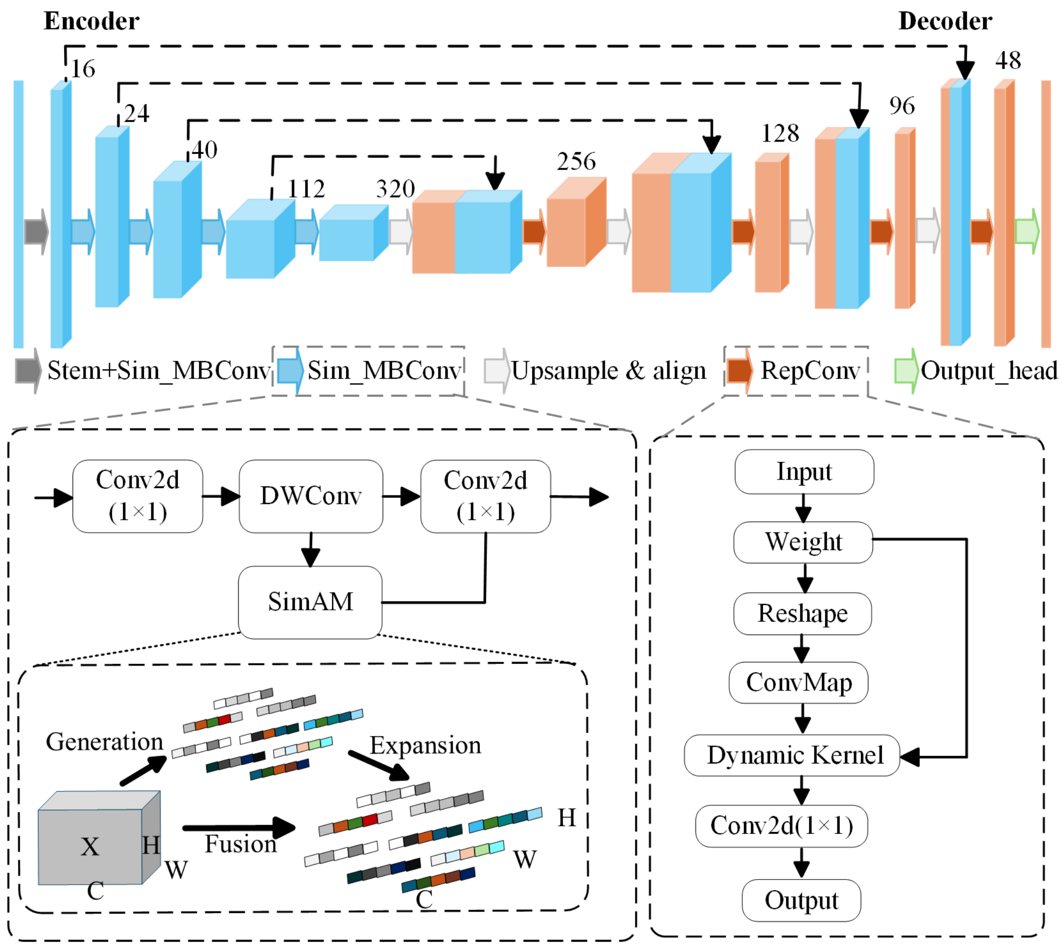

Fig. 3. Architecture of SimEffB0-RepUNet, which uses a SimAM-enhanced EfficientNet-B0 encoder and RepConv modules in the decoder.

Experimental design

Leaf segmentation

For leaf segmentation, U-Net, DeepLabV3+, PSPNet, HRNet, and SegFormer were compared using mPA, IoU-Background, IoU-Leaf, mIoU, parameter count, and inference time. IoU-Leaf was treated as the primary metric. Backbone selection was then evaluated within DeepLabV3+ using MobileNetV2, MobileNetV3, MobileNetV4, StarNetS1, and EfficientNet-B0. Finally, MobileNetV3 with SE attention was compared with the SimAM-enhanced variant to assess the contribution of parameter-free spatial attention.

Lesion segmentation

For lesion segmentation, candidate architectures were first compared to select the baseline, followed by ablation experiments on backbone, attention module, and decoder convolution. The evaluated backbones included VGG16, ResNet50, ResNetRS50, MobileNetV2, MobileNetV3, and EfficientNet-B0. The attention comparison included SE, ECA, CBAM, and SimAM, and the decoder comparison included Conv, Conv×2, RepConv, ODConv, PConv, and CondConv2D. IoU-Disease was the primary metric, together with mPA, IoU-Background, mIoU, parameter count, and inference time.

Model Comparison Experiments

To evaluate the dual-stage strategy, five single-stage models were compared with a conventional DeepLabV3+ + U-Net pipeline and the proposed SimV3-DL3+ + SimEffB0-RepUNet pipeline. Performance was measured by IoU-Leaf, IoU-Disease, parameter count, and average inference time per image. For dual-stage models, inference time included both segmentation stages and intermediate processing.

Expert Visual Evaluation and Model Prediction

To compare automated severity estimation with human judgment, 15 trained technicians visually estimated disease severity on representative leaf images. Their assessments were compared with model predictions for the same images.

Experimental setup

Experiments were conducted on an NVIDIA GeForce RTX 4060 Ti GPU with an Intel Core i5-13490F CPU and 32 GB RAM. Models were trained in Python 3.8 with CUDA 11.8 using Adam, a batch size of 2, an initial learning rate of 1×10^{-5}, cosine annealing to 1×10^{-6}, and 500 epochs. All images were resized to 512×512. Dice loss was used:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2|X \cap Y|}{|X| + |Y|} \tag{1}$$

where X and Y denote the predicted and ground-truth masks.

Evaluation metrics

Segmentation performance was evaluated using PA, mPA, IoU, and mIoU:

$$PA = \frac{\sum_{i=0}^{k} P_{\mathrm{ii}}}{\sum_{i=0}^{k} \sum_{j=0}^{k} P_{\mathrm{ij}}} \tag{2}$$

where P_{ii} denotes correctly classified pixels of class i, and P_{ij} denotes pixels whose ground-truth class is i but are predicted as class j.

$$\mathrm{mPA} = \frac{1}{K+1} \sum_{i=0}^{K} \frac{P_{\mathrm{ii}}}{\sum_{j=0}^{K} P_{\mathrm{ij}}} \tag{3}$$

Here, $K+1$ is the number of classes including background.

$$\mathrm{IoU}_i = \frac{TP_i}{TP_i + FP_i + FN_i} \tag{4}$$

where TP_i, FP_i, and FN_i are true-positive, false-positive, and false-negative pixels for class i.

$$\mathrm{mIoU} = \frac{1}{K+1} \sum_{i=0}^{K} \mathrm{IoU}_i = \frac{1}{K+1} \sum_{i=0}^{K} \frac{TP_i}{TP_i + FP_i + FN_i} \tag{5}$$

Severity estimation was evaluated using R^2, MAE, and RMSE:

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \tilde{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2} \tag{6}$$

where n is the number of samples, y_i and $\tilde{y}_i$ are the actual and predicted values, and $\bar{y}$ is the mean of all actual values.

$$\mathrm{MAE} = \frac{1}{n} \sum_{i=1}^{n} |y_i - \tilde{y}_i| \tag{7}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \tilde{y}_i)^2} \tag{8}$$

Results

Leaf segmentation

Among the five leaf segmentation models (Fig. 4), DeepLabV3+ achieved the best overall accuracy, with an mPA of 99.11%, IoU-Background of 98.70%, IoU-Leaf of 97.49%, and mIoU of 98.09%. It also maintained moderate model size and inference time, and was therefore selected as the baseline for leaf segmentation.

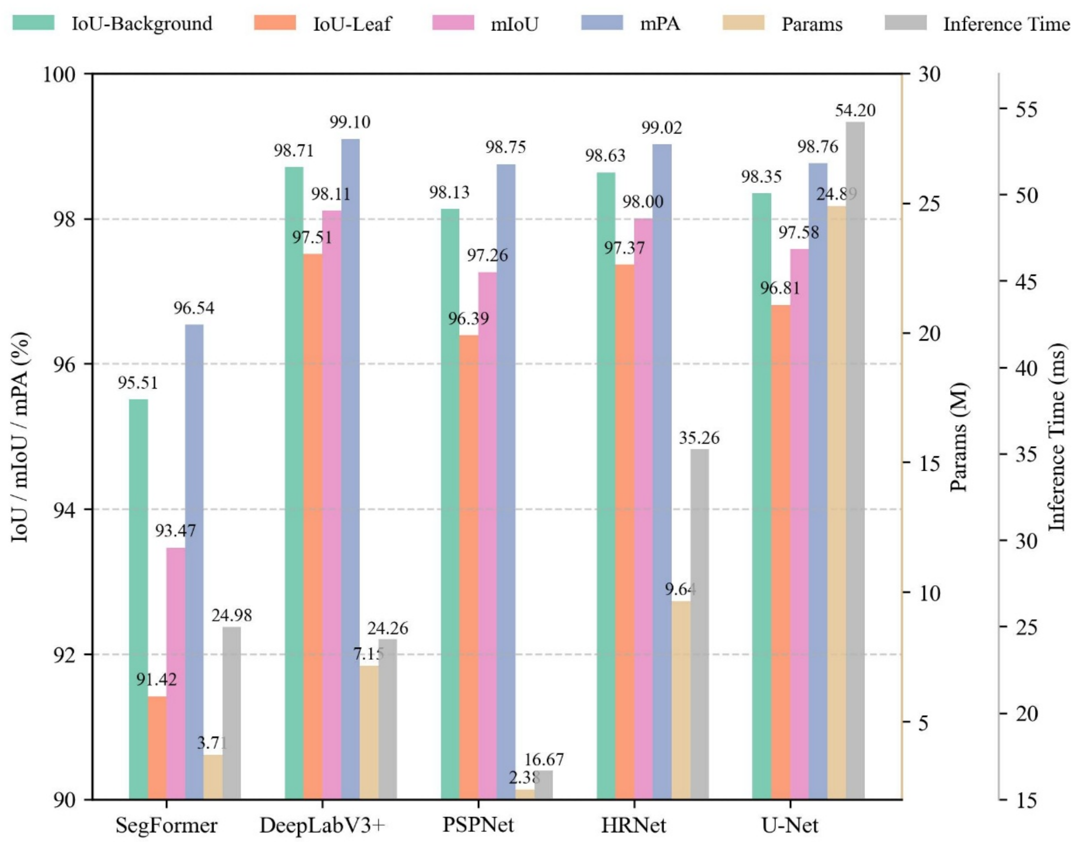

Fig. 4. Leaf segmentation comparison of SegFormer, DeepLabV3+, PSPNet, HRNet, and U-Net.

Backbone comparison further showed that Sim_MobileNetV3 provided the best accuracy–efficiency balance (Fig. 5), reaching the highest mPA (99.19%) and mIoU (98.30%) with only 4.83 M parameters and 20.62 ms inference time.

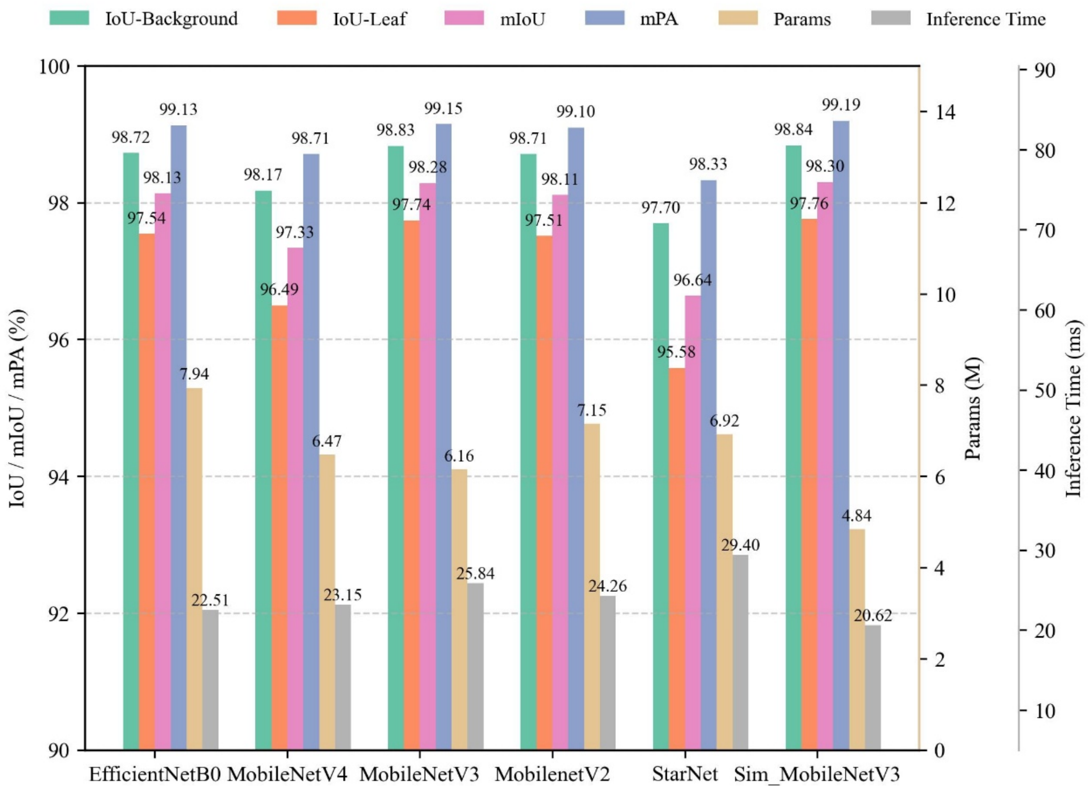

Fig. 5. Leaf segmentation performance of DeepLabV3+ with different backbones.

Lesion segmentation

For lesion segmentation, U-Net achieved the highest accuracy among the candidate architectures (Fig. 6), with an mIoU of 84.97% and IoU-Disease of 73.07%. Although it had the largest parameter count (24.89 M) and longest inference time (51.53 ms), its accuracy made it the baseline for subsequent optimization.

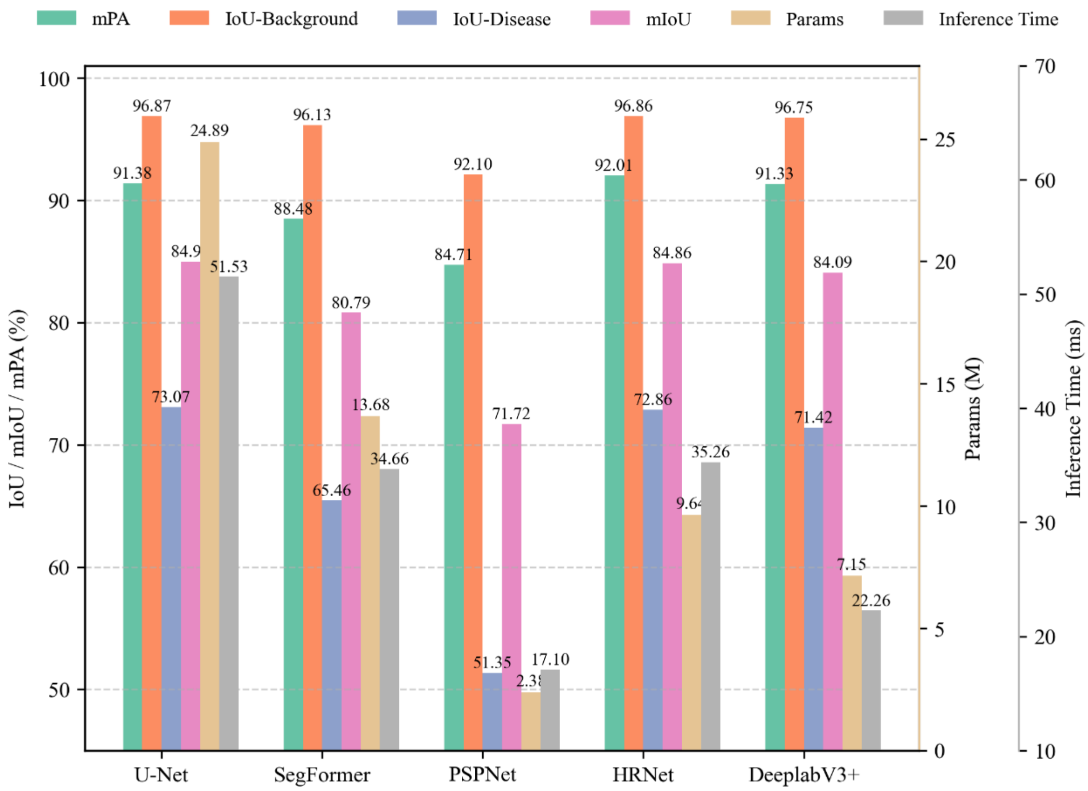

Fig. 6. Lesion segmentation comparison of U-Net, SegFormer, PSPNet, HRNet, and DeepLabV3+.

Backbone evaluation showed that EfficientNet-B0 achieved the best balance for lesion segmentation (Fig. 7), providing the highest mIoU and IoU-Disease while reducing parameters by more than 82% compared with ResNet50 and achieving the fastest inference time (25.01 ms).

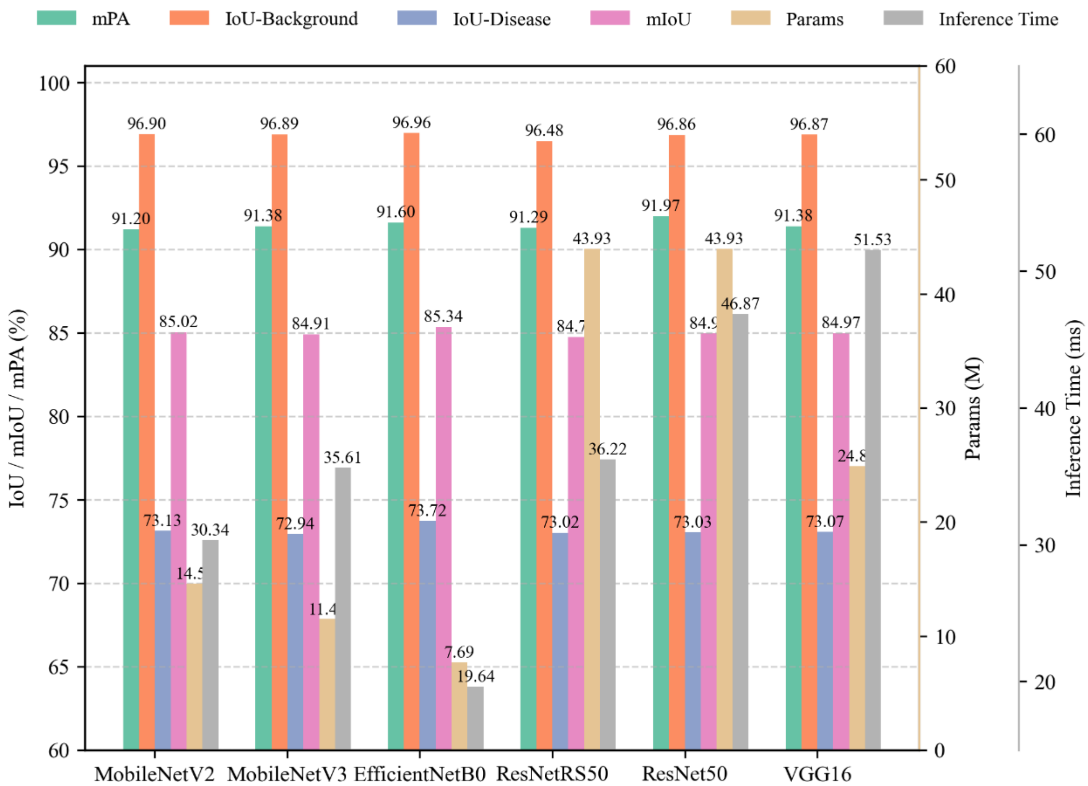

Fig. 7. Lesion segmentation performance of U-Net with different backbones.

Attention comparison indicated that SimAM improved mIoU (85.46%) and IoU-Disease (73.94%) over SE while reducing the parameter count to 7.05 M (Fig. 8). It therefore provided the most favorable trade-off among the evaluated attention modules.

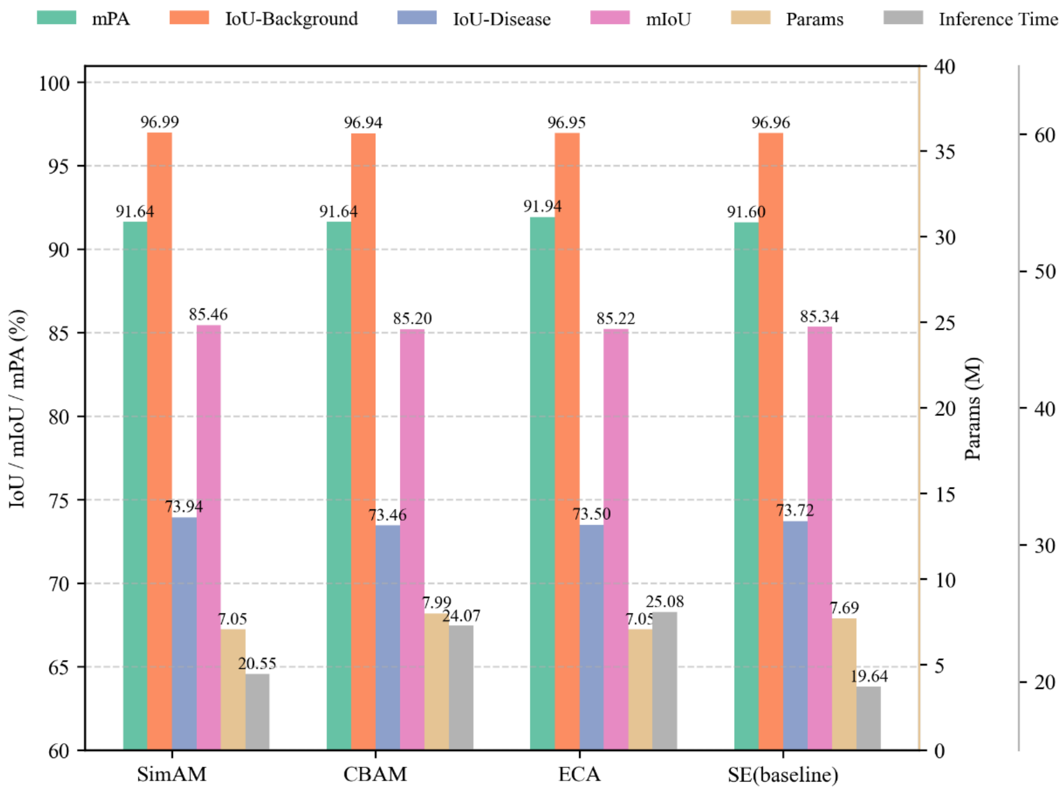

Fig. 8. Lesion segmentation performance with different attention modules.

Among decoder variants, RepConv achieved the best overall performance (Fig. 9), with the highest IoU-Disease (74.30%) and mIoU (85.64%), a moderate parameter count (7.73 M), and the shortest inference time (17.33 ms). RepConv was therefore adopted in the final lesion model.

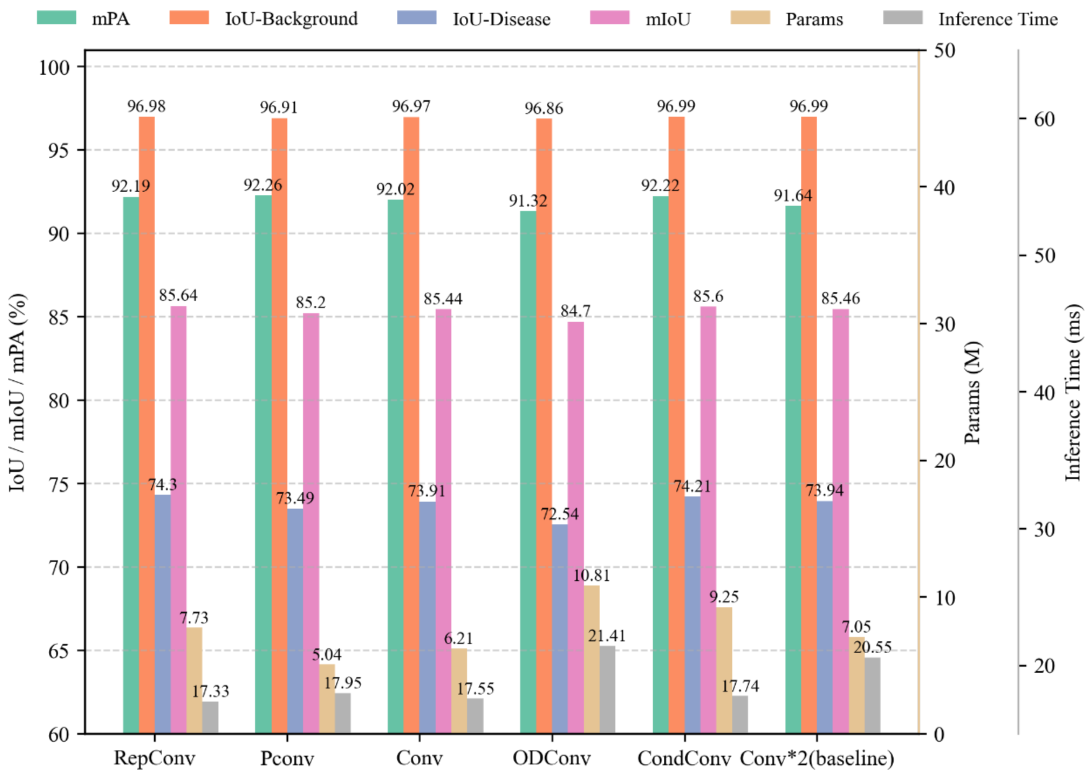

Fig. 9. Lesion segmentation performance with different decoder convolution variants.

Severity estimation based on SimEffB0-RepUNet outputs strongly agreed with ground truth (Fig. S1), achieving $R^2 = 0.976$, MAE = 0.012, and RMSE = 0.019 on the test set.

Model comparison

The proposed dual-stage model outperformed both single-stage models and the conventional dual-stage baseline (Fig. 10). It achieved 97.76% IoU-Leaf and 74.30% IoU-Disease with 12.56 M parameters and 2261 ms inference time. In comparison, DeepLabV3+ + U-Net reached 97.49% IoU-Leaf and 72.71% IoU-Disease with 32.04 M parameters and 2303 ms inference time. Qualitative results (Fig. S2) further showed that the proposed model reduced leaf over-segmentation and improved lesion boundary delineation.

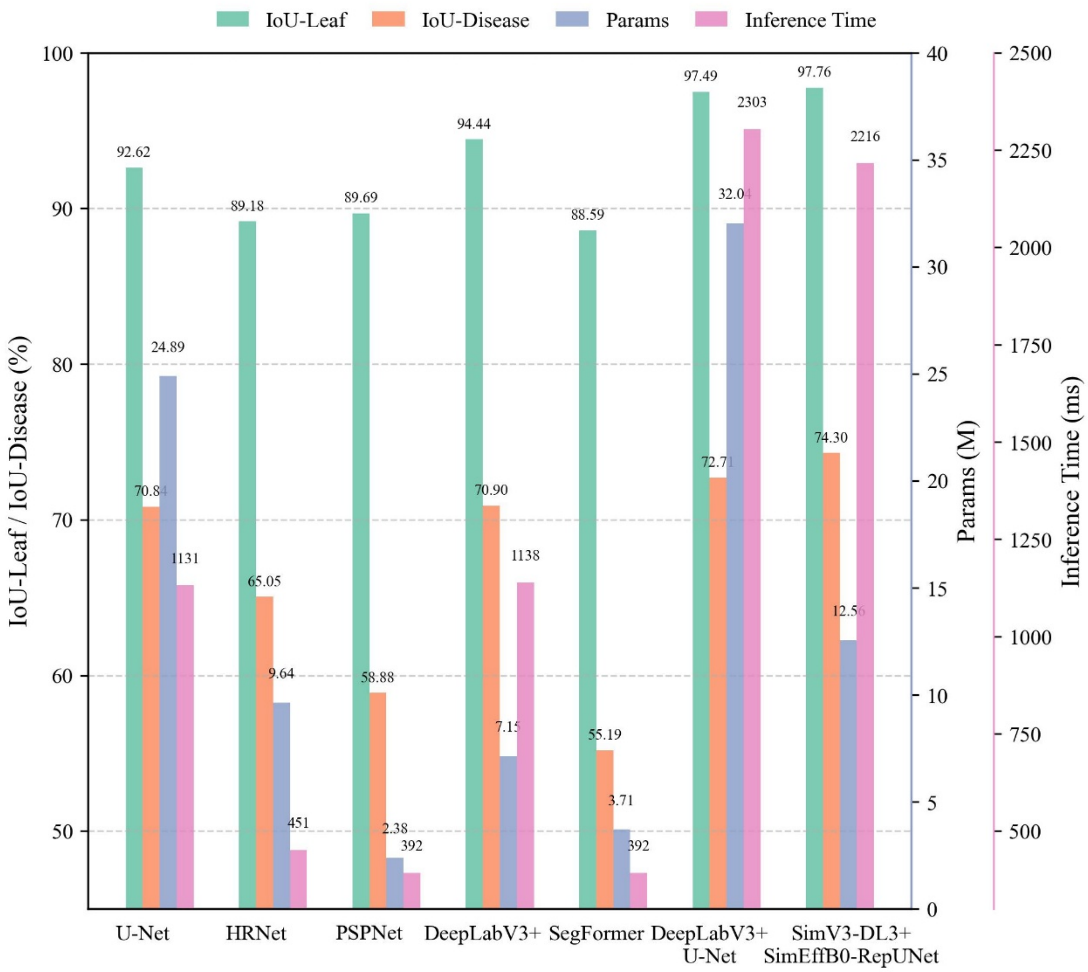

Fig. 10. Performance comparison of single-stage and dual-stage segmentation models.

Expert Visual Evaluation and Model Prediction

Expert visual estimates showed clear inter-observer variability (Fig. S3). For a mildly diseased leaf, the mean expert estimate was 4.56%, whereas the model predicted 2.32%. For a more severely diseased leaf, the mean expert estimate (20.01%) was close to the model prediction (19.32%). These results indicate that automated estimation can reduce subjectivity, particularly under mild disease conditions.

Discussion

The results demonstrate that separating leaf and lesion segmentation is effective for disease severity quantification under field conditions. The first stage provides a reliable leaf area for severity calculation, while the second stage focuses on small and irregular symptomatic regions. Compared with single-stage models, this task decomposition reduces feature interference and improves the stability of lesion delineation and severity estimation.

Attention-guided lightweight leaf segmentation for robust field conditions

The leaf segmentation results show that DeepLabV3+ with a SimAM-enhanced MobileNetV3 backbone improves boundary extraction while keeping the model compact. SimAM enhances spatial sensitivity without adding parameters, helping suppress background interference and reduce over-segmentation. This is important because errors in leaf area directly propagate to severity estimation.

Lesion segmentation model performance enhancement

For lesion segmentation, the performance gains arise from the combined use of EfficientNet-B0, SimAM, and RepConv. EfficientNet-B0 provides a compact encoder for multi-scale lesion features, SimAM improves spatial localization of small symptoms, and RepConv strengthens boundary reconstruction during training while preserving inference efficiency. Together, these components improve IoU-Disease and mIoU without the large parameter cost of heavier U-Net variants.

Performance of the dual-stage model

The comparison with single-stage and conventional dual-stage models confirms that independent optimization of leaf and lesion segmentation is beneficial. The proposed framework improves both leaf and lesion IoU while using fewer parameters than the DeepLabV3+ + U-Net baseline. This also explains the strong agreement between predicted and reference severity values.

Practical deployment and applications

To support field use, the model was integrated into a Flutter-based Android application (https://github.com/husnxjsjsb/Grape-Downy). This implementation demonstrates the feasibility of deploying the proposed framework for portable disease monitoring and decision support.

Limitations

Several limitations remain. DSLL-Net uses two networks and is therefore slower than some single-stage alternatives. The current evaluation focuses on one crop-disease system, so broader validation across crops, diseases, sensors, and field environments is needed. Pixel-level annotation is also labor-intensive; semi-supervised or weakly supervised learning may reduce this cost. Finally, severity is estimated from lesion area alone, and future work may incorporate temporal or multispectral information.

Conclusion

This study proposes DSLL-Net, a lightweight dual-stage framework for grapevine downy mildew severity assessment. SimV3-DL3+ extracts leaf regions from field backgrounds, while SimEffB0-RepUNet segments small and irregular lesions using a SimAM-EfficientNet-B0 encoder and RepConv decoder. The final model achieved 74.3% IoU-Disease with 12.56 M parameters and produced severity estimates highly consistent with reference values ($R^2 = 0.976$). These results indicate that DSLL-Net provides an efficient and deployable solution for quantitative disease monitoring. Future work will expand validation across more crops, diseases, and field conditions and further optimize edge-device deployment.

Data and code availability

The data and code used in this study are available on GitHub (https://github.com/).

Author contribution statement

Changqing Yan, Gang Zhao: Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. Shuyang Li: Writing – original draft, Visualization, Validation, Methodology, Data curation, Conceptualization. Ling Yin: Supervision, project administration, funding acquisition. Zhe Liu, Shumei Wei, Xin Lin, Xin Li, Jindong Liu, Zhenpeng Huang, Danni Zheng, Zeyun Liang, Linjia Yao, Bin Chen, Qi Tian: Data curation, Investigation, Validation, Writing – review & editing. Chenliang Wang, Junjie Qu: Writing – review & editing, Visualization, Validation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary Materials

Figs.S1-Figs.S4: Fig.S1 Fig.S2 Fig.S3 Fig.S4.

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